

1.3.5 – 1.3.6 TCP/IP Model vs OSI Model

ٹی سی پی / آئی پی (TCP/IP) ماڈل حقیقی انٹرنیٹ پر استعمال ہونے والا عملی ماڈل ہے اور اس کی صرف چار تہیں ہیں۔

CS XI

Chapter 1: Computer Systems

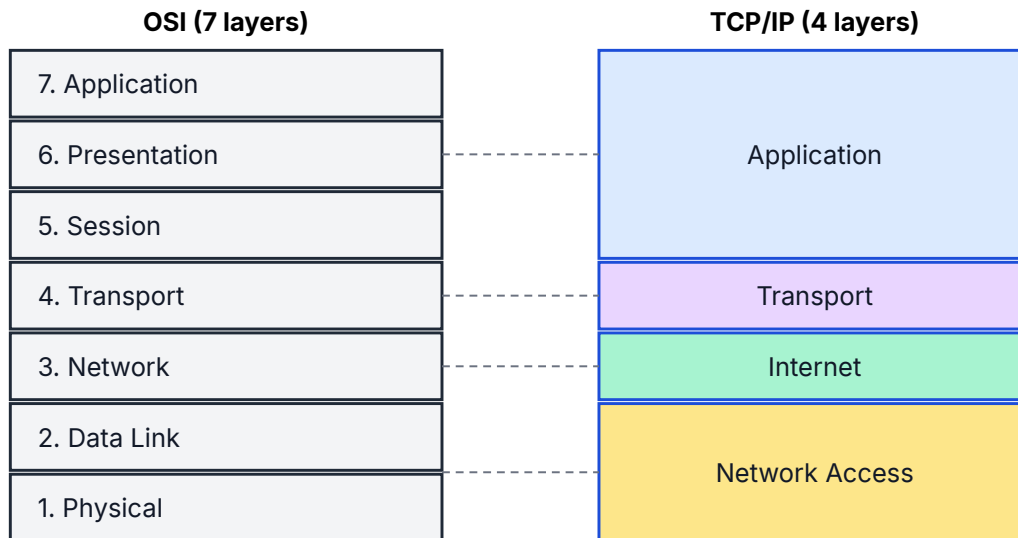
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LEARN IT

The OSI model is mainly a **reference** model used for teaching and design. The **TCP/IP model** is the **practical model used on the real Internet**. It condenses the 7 OSI layers into just **4 layers**.

TCP/IP model (4 layers)	Corresponds to OSI layers
1. Application	Application, Presentation, Session (7, 6, 5)
2. Transport	Transport (4)
3. Internet	Network (3)
4. Network Access	Data Link, Physical (2, 1)



How the 7 OSI layers map onto the 4 TCP/IP layers.

Point	OSI model	TCP/IP model
Number of layers	7	4
Use	Reference / teaching model	Practical model used on the real Internet
Top layers	Separate Application, Presentation and Session layers	All three combined into one Application layer
Bottom layers	Separate Data Link and Physical layers	Combined into one Network Access layer

WORKED EXAMPLE

Where does each protocol/device go in TCP/IP?

HTTP (OSI layer 7) → TCP/IP **Application**.

SSL/TLS (OSI layer 6) → TCP/IP **Application** (layers 7, 6, 5 are merged).

UDP (OSI layer 4) → TCP/IP **Transport**.

IP and routers (OSI layer 3) → TCP/IP **Internet**.

Ethernet, switches, cables (OSI layers 2 and 1) → TCP/IP **Network Access**.

REMEMBER IT

Count it as **3-1-1-2**: the TCP/IP Application layer takes 3 OSI layers, Transport takes 1, Internet takes 1, Network Access takes 2. Total $3+1+1+2 = 7$.

COMMON MISTAKE

The OSI "Network" layer (3) matches the TCP/IP "**Internet**" layer, not the TCP/IP "Network Access" layer. The similar names confuse many students.

اردو میں سمجھیں

ٹی سی پی/آئی پی (TCP/IP) ماڈل حقیقی انٹرنیٹ پر استعمال ہونے والا عملی ماڈل ہے اور اس کی صرف چار تہیں ہیں۔ اس کی ایپلی کیشن تہ OSI کی 7، 6 اور 5 تہوں کے برابر ہے، ٹرانسپورٹ تہ 4 کے، انٹرنیٹ تہ 3 کے، اور نیٹ ورک ایکسیس تہ 2 اور 1 کے برابر ہے۔

CHECK YOURSELF

1. How many layers does the TCP/IP model have?
2. Which TCP/IP layer corresponds to OSI layers 2 and 1?
3. Which TCP/IP layer contains the OSI Session layer?

Answers: 1. Four. 2. Network Access. 3. Application.

Next lecture: Chapter review (in the chapter lecture PDF)